

Book Reviews

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Applied Computational Aerodynamics. A Modern Engineering Approach

Russell M. Cummings, William H. Mason, Scott A. Morton, and David R. McDaniel, Cambridge University Press, New York, 2015, 849 pp., \$125.00

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The application of computational fluid dynamics to aerodynamic design has become a mature field where well-established codes are used for many of the real-world applications. Although there is always going to be research related to algorithm and code development or advanced physics modeling, an important aspect of the work of future aerospace engineers is toward product development and integration using existent codes. Thus, the book aims at addressing a paradigm shift in undergraduate academic curriculum from developers of computational aerodynamics tools toward “intelligent users” of computational aerodynamics.

The text is targeted at advanced undergraduates in aerospace engineering learning computational aerodynamics while also learning engineering methods and aerodynamics. Even though relatively long, the text has an easy style, with useful quotes, insets containing biographies of people who work in the field, and concept boxes that summarize the most important ideas. There are also projects included for most chapters, together with traditional problems. These features definitely make the text captivating and should help undergraduate students become passionate about the field.

The book is organized into 10 chapters, with an addition of six appendices. The first two chapters introduce the computational aerodynamics concepts and process, from geometry description to flow solution and to postprocessing in the broader context of experimental, theoretical, and engineering methods and, using the historical perspective, in connection with the evolution of computers. Chapter 3 contains the derivation of the governing equations together with mathematical classifications and several inviscid and viscous flow simplifications useful for aerodynamics, such as potential flow models and boundary-layer equations. In my opinion, this chapter could have tried to better differentiate the compressible and incompressible cases. Boundary conditions, which are different for the two cases, as well as the numerical methods, would be better understood in this

context. Also, the book would have benefited from a discussion on shock waves from the point of view of the governing equations. This would have added to the scope of the book and would have increased the audience.

Chapter 4 introduces the notions of airfoil and wing, starting from potential flow theory and basic aerodynamics. The necessary background on transonic, supersonic, and hypersonic theories is also presented. The next three chapters discuss computational methods for aerodynamics. Thus, Chapter 5 presents the classical linear theory and its applications in panel methods and vortex lattice methods. Chapter 6 provides background on finite differences, finite volume, finite element, and pseudospectral methods, as well as time integration. As required for a computational aerodynamics book, the wave, heat, and Laplace equations are all covered. Since dealing with complex boundaries is an important part of aerodynamics, Chapter 7 covers computational grid features, including types, generation, and grid adaptation. The book continues, with Chapter 8 dedicated to turbulence modeling and Chapter 9 on visualization. I found this chapter, which is lacking in most other aerodynamics books, especially useful from the point of view of the practitioner. The text helps the student get familiar with various visualization techniques used not only by computational aerodynamicists but also by experimentalists. Finally, Chapter 10 provides an integration of the notions introduced in the book to actual airplane design concepts. The appendices contain very useful information on open-source aerodynamics codes, turbulence models, and code validation against experimental data.

In summary, I really like the practical emphasis of the presentation and recommend it as an excellent material for an undergraduate class in computational aerodynamics.

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